Inequalities of Real Numbers.

Bernoulli's Inequality

If $a>-1$, $1+na\leq(1+a)^n$ for all $n\in\mathbb{N}$.

Proof. If $n=1$, trivial. If the statement hold for $n=k\in\mathbb{N}$, then

$$(1+a)^{k+1}=(1+a)(1+a)^k \underset{1+a\geq 0}{\geq}(1+a)(1+ka)=1+(k+1)a+a^2\geq 1+(k+1)a.$$

Arithmetic-Geometric mean

If $a,b\in\mathbb{R}$, then $2ab\leq a^2+b^2$ (Directly, it can be proved by $(a-b)^2\geq 0$.)

Generalized A-G mean inequality,

If $a_i\geq 0$, then for all $n\in\mathbb{N}$,

$$(a_1\cdot a_2\cdots a_n)^{\frac{1}{n}}\leq \frac{1}{n}\cdot(a_1+a_2+\cdots+a_n)$$

proof. If $n=2^1$, $\sqrt{a_1a_2}\leq\frac{1}{2}(a_1+a_2)$, thus the statement holds.
Suppose that for $n=2^k$ the statement be hold. That is,

$$(a_1\cdots a_n)^{\frac{1}{n}}\leq\frac{1}{n}(a_1+\cdots+a_n)$$

Now,
$$\begin{aligned}\frac{1}{2n}(a_1+\cdots+a_{2n})&=\frac{1}{2}\left(\frac{1}{n}(a_1+\cdots+a_n)+\frac{1}{n}(a_{n+1}+\cdots+a_{2n})\right)\\
&\geq\frac{1}{2}\left((a_1\cdots a_n)^{\frac{1}{n}}+(a_{n+1}\cdots a_{2n})^{\frac{1}{n}}\right)\\
&=\frac{1}{2}\left(\left((a_1\cdots a_n)^{\frac{1}{2n}}\right)^2+\left((a_{n+1}\cdots a_{2n})^{\frac{1}{2n}}\right)^2\right)\\
&\geq\frac{1}{2}\cdot 2(a_1\cdots a_n)^{\frac{1}{2n}}\cdot(a_{n+1}\cdots a_{2n})^{\frac{1}{2n}}\\
&=(a_1\cdots a_na_{n+1}\cdots a_{2n})^{\frac{1}{2n}}\end{aligned}$$

Therefore, by induction, the statement hold when $n=2^k$ $(k\in\mathbb{N})$.

To generalize, let $n\geq 2$ be given. Put $k\in\mathbb{N}$ s.t

$$2^{k-1}\leq n\leq 2^k \qquad \frac{1}{2^k}\leq\frac{1}{n}\leq\frac{1}{2^{k-1}} \qquad 1\leq 2$$

$$\frac{n}{2^k}\leq 1$$

$$0\leq 1-\frac{n}{2^k}$$

Put

$$b_i=\begin{cases}a_i & \text{if } i\leq n\\ \frac{1}{n}(a_1+\cdots+a_n)(=M) & \text{if } n<i\leq 2^k\end{cases}$$

Then, $(b_1\cdots b_{2^k})^{\frac{1}{2^k}}\leq\frac{1}{2^k}\cdot(b_1+\cdots+b_{2^k})=\frac{1}{2^k}\cdot(nM+(2^k-n)M)=M$

And $(b_1\cdots b_{2^k})^{\frac{1}{2^k}}=(a_1\cdots a_n\cdot M^{2^k-n})^{\frac{1}{2^k}}=(a_1\cdots a_n)^{\frac{1}{2^k}}\cdot M^{(1-\frac{n}{2^k})}$

Finally, $(a_1\cdots a_n)^{\frac{1}{2^k}}\leq M^{\frac{n}{2^k}}$ implies $(a_1\cdots a_n)^{\frac{1}{n}}\leq(a_1+\cdots+a_n)\cdot\frac{1}{n}$.

Lagrange's Inequality.

Let $a_i, b_i \in \mathbb{R}$. Then,

$$\left[\sum_{i=1}^{n} a_i b_i\right]^2 = \left[\sum_{i=1}^{n} a_i^2\right] \cdot \left[\sum_{i=1}^{n} b_i^2\right] - \frac{1}{2} \cdot \sum_{1 \le j,k \le n} (a_j b_k - a_k b_j)^2$$